

Complete 14-Bit Nanoprocessor - Opcodes & Flags Guide

This is the complete reference guide for the extended 14-bit nanoprocessor.

Part 1: 14-Bit Instruction Format

Bit position: 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Field	Bits	Description
OPCODE	13-10	Instruction type (4 bits)
R1	9-7	First register (destination/source)
R2	6-4	Second register (source)
FUNC/IMM	3-0	Function code or Immediate value

Part 2: Complete Opcode Table

Opcode	Hex	Instruction	Operation	Flags Affected
0000	0	MOVI	R1 = Immediate	None
0001	1	ADD	R1 = R1 + R2	Zero, Overflow
0010	2	SUB	R1 = R1 - R2	Zero, Overflow
0011	3	NEG	R1 = -R1	Zero, Overflow
0100	4	JZR	If R1=0 then PC=d	None
0101	5	IN	R1 = Switches[3:0]	None
0110	6	MUL	R1 = (R1 * R2) mod 16	Zero, Overflow
0111	7	AND	R1 = R1 AND R2	Zero
1000	8	OR	R1 = R1 OR R2	Zero
1001	9	XOR	R1 = R1 XOR R2	Zero
1010	A	NOT	R1 = NOT R1	Zero
1011	B	CMP	Compare R1 and R2	CMP_Equal, CMP_Less, CMP_Greater

Part 3: Each Instruction in Detail

1. MOVI (Move Immediate) - Opcode 0000

Syntax: MOVI R1, d

Operation: $R1 = d$

Format: 0000 R1 000 dddd

Example: MOVI R7, 5 -> 0x0E05

2. ADD (Add) - Opcode 0001

Syntax: ADD R1, R2

Operation: $R1 = R1 + R2$

Format: 0001 R1 R2 0000

Example: ADD R7, R1 -> 0x1E10

Flags: Zero, Overflow

3. SUB (Subtract) - Opcode 0010

Syntax: SUB R1, R2

Operation: $R1 = R1 - R2$

Format: 0010 R1 R2 0000

Example: SUB R7, R1 -> 0x2E10

Flags: Zero, Overflow

4. NEG (Negate) - Opcode 0011

Syntax: NEG R1

Operation: $R1 = -R1$ (2's complement)

Format: 0011 R1 000 0000

Example: NEG R7 -> 0x3E00

Flags: Zero, Overflow

5. JZR (Jump if Zero) - Opcode 0100

Syntax: JZR R1, d

Operation: If $R1 = 0$ then $PC = d$ else $PC = PC + 1$

Format: 0100 R1 000 0ddd

Example: JZR R0, 4 -> 0x4004

6. IN (Input from Switches) - Opcode 0101

Syntax: IN R1

Operation: $R1 = \text{Switches}[3:0]$

Format: 0101 R1 000 0000

Example: IN R7 -> 0x5E00

7. MUL (Multiply) - Opcode 0110

Syntax: MUL R1, R2

Operation: $R1 = (R1 * R2) \bmod 16$

Format: 0110 R1 R2 0000
Example: MUL R1, R2 -> 0x60A0
Flags: Zero, Overflow

8. AND (Bitwise AND) - Opcode 0111

Syntax: AND R1, R2
Operation: R1 = R1 AND R2
Format: 0111 R1 R2 0000
Example: AND R1, R2 -> 0x70A0
Flags: Zero

9. OR (Bitwise OR) - Opcode 1000

Syntax: OR R1, R2
Operation: R1 = R1 OR R2
Format: 1000 R1 R2 0000
Example: OR R1, R2 -> 0x80A0
Flags: Zero

10. XOR (Bitwise XOR) - Opcode 1001

Syntax: XOR R1, R2
Operation: R1 = R1 XOR R2
Format: 1001 R1 R2 0000
Example: XOR R1, R2 -> 0x90A0
Flags: Zero

11. NOT (Bitwise NOT) - Opcode 1010

Syntax: NOT R1
Operation: R1 = NOT R1
Format: 1010 R1 000 0000
Example: NOT R1 -> 0xA100
Flags: Zero

12. CMP (Compare) - Opcode 1011

Syntax: CMP R1, R2
Operation: Compare R1 and R2, set flags, NO register write!
Format: 1011 R1 R2 0000
Example: CMP R1, R2 -> 0xB0A0
Flags: CMP_Equal, CMP_Less, CMP_Greater

Part 4: Flag System

Flag	LED	Pin	Meaning
Data[0]	LD0	U16	R7 bit 0 (LSB)
Data[1]	LD1	E19	R7 bit 1

Data[2]	LD2	U19	R7 bit 2
Data[3]	LD3	V19	R7 bit 3 (MSB)
CMP_Equal	LD8	V13	1 when compared values equal
CMP_Less	LD9	V3	1 when R1 < R2
CMP_Greater	LD10	W3	1 when R1 > R2
Zero Flag	LD14	P1	1 when ALU result = 0
Overflow Flag	LD15	L1	1 when result outside -8 to +7

Part 5: Switch Mapping

SW0-SW3: Data input / Immediate value

SW4-SW6: Register R1

SW7-SW9: Register R2

SW10-SW13: Opcode

SW14: 0=Register Mode, 1=Immediate Mode

SW15: 0=ROM Mode, 1=Manual Mode

Part 6: Button Functions

BTNC: Reset

BTNR: Execute/Store

BTNL: Toggle display between R7 and Input

Part 7: Register Reference

Register	Binary	Special Function
R0	000	Read-only, always 0
R1-R6	001-110	General purpose
R7	111	Displayed on 7-segment & LEDs

Part 8: Example Programs

Example 1: Count from 0 to 7

MOVI R7, 0

MOVI R1, 1

ADD R7, R1 (Repeat)

JZR R0, 2